IGCSE 0607 Practice Questions — E6.4

Surface area and volume

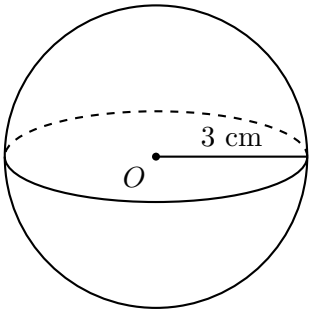
Total Questions: 9 — **Total Marks:** 67 Marks

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  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

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1 The diagram shows a solid sphere with center O and radius 3 cm.



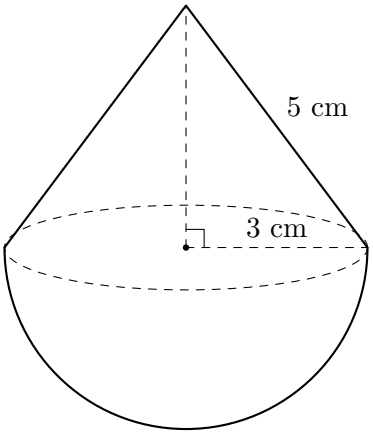
(a) Calculate the exact volume of the sphere.
Give your answer in terms of π.

..... cm^3 **[2]**

(b) Calculate the exact surface area of the sphere.
Give your answer in terms of π.

..... cm^2 **[2]**

2 The diagram shows a solid formed by joining a cone to a hemisphere with the same radius.
The radius of the hemisphere is 3 cm and the slant height of the cone is 5 cm.



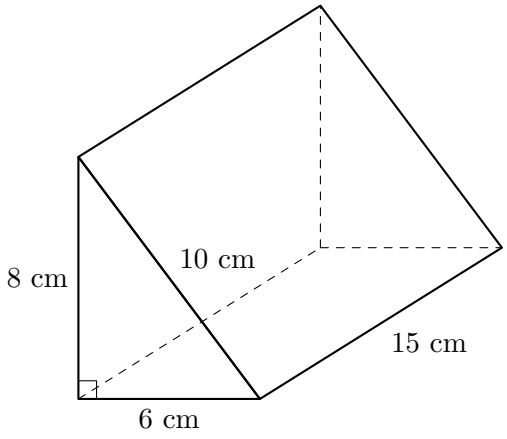
(a) Calculate the exact volume of the solid.
Give your answer in terms of π.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.
Give your answer in terms of π.

..... cm^2 **[3]**

3 The diagram shows a solid triangular prism.
The cross-section of the prism is a right-angled triangle with sides 6 cm, 8 cm and 10 cm.
The length of the prism is 15 cm.



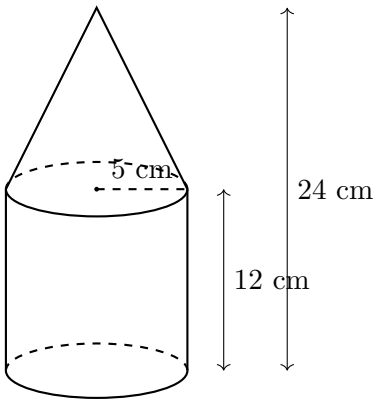
(a) Calculate the volume of the prism.

..... cm^3 **[2]**

(b) Calculate the total surface area of the prism.

..... cm^2 **[3]**

4 A solid is formed by attaching a right circular cone to the top of a solid cylinder.
The cylinder and the base of the cone both have a radius of 5 cm.
The height of the cylinder is 12 cm and the total height of the solid is 24 cm.



(a) Calculate the volume of the solid.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[5]**

5 A solid metal cylinder has radius x cm and height $3x$ cm.
The cylinder is melted down and completely recast into solid spheres.
Each sphere has a radius of $\frac{x}{2}$ cm.

(a) Find an expression, in terms of x and π, for the volume of the cylinder.
Give your answer in its simplest form.

..... cm^3 **[2]**

(b) Find an expression, in terms of x and π, for the volume of one sphere.
Give your answer in its simplest form.

..... cm^3 **[2]**

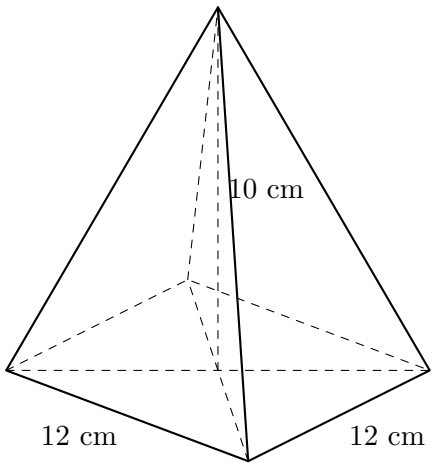
(c) Work out the number of complete spheres that can be made.

..... **[1]**

(d) The total surface area of the original cylinder is 140 cm^2.
Work out the value of x.

$x =$ ..... **[3]**

6 A solid metal pyramid has a square base of side 12 cm and a perpendicular height of 10 cm.



(a) Calculate the volume of the pyramid.

..... cm^3 **[2]**

(b) Calculate the total surface area of the pyramid.

..... cm^2 **[4]**

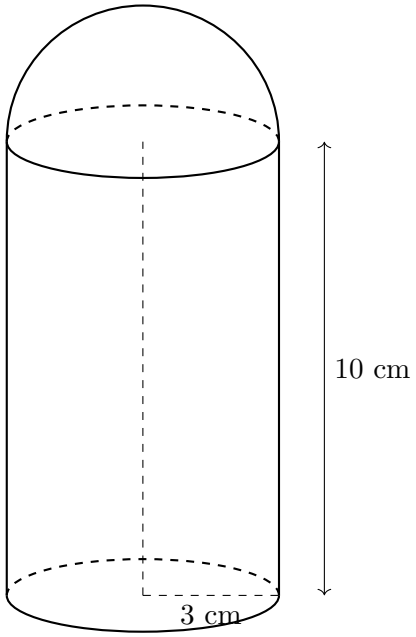
(c) The metal pyramid is melted down and recast to make solid spheres of radius 2.5 cm.
Calculate the number of complete spheres that can be made.

..... **[3]**

(d) A mathematically similar pyramid has a volume of 1620 cm^3.
Calculate the perpendicular height of this similar pyramid.

..... cm **[3]**

7 A solid is made by joining a solid hemisphere of radius 3 cm to a solid cylinder of radius 3 cm and height 10 cm.



NOT TO SCALE

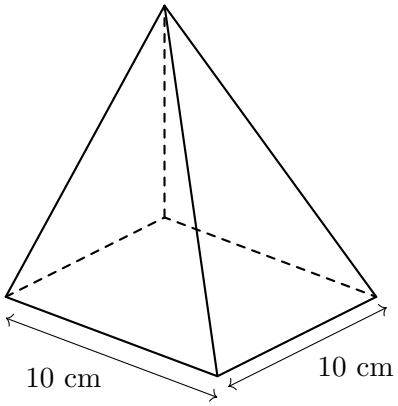
(a) Work out the volume of the solid.
Give your answer in terms of π.

..... cm^3 **[3]**

(b) Work out the total surface area of the solid.
Give your answer in terms of π.

..... cm^2 **[3]**

8 The diagram shows a solid pyramid with a square base.
The base has sides of length 10 cm.
The volume of the pyramid is 400 cm^3.



NOT TO SCALE

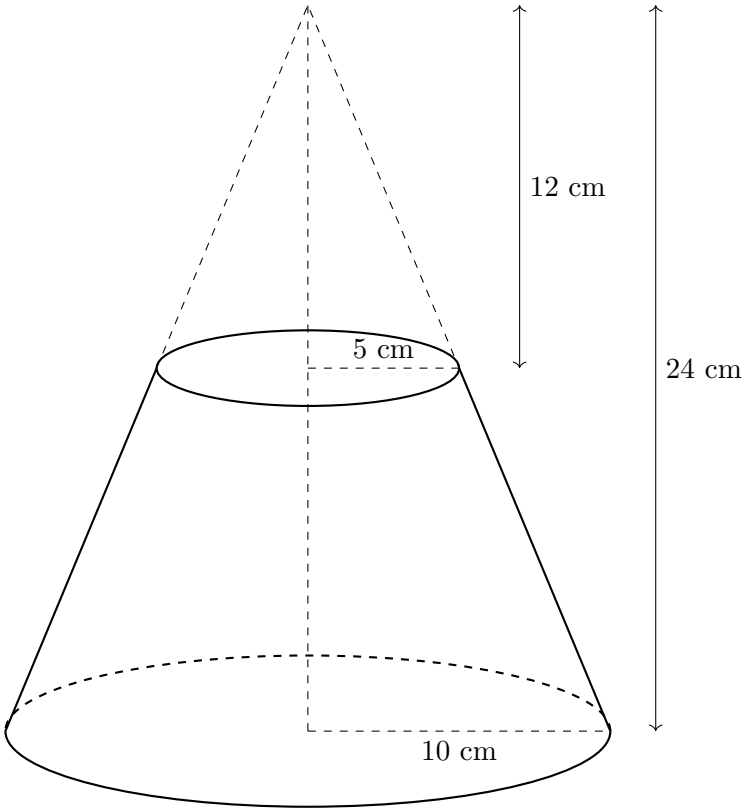
(a) Calculate the perpendicular height of the pyramid.

..... cm **[2]**

(b) Calculate the total surface area of the pyramid.

..... cm^2 **[4]**

9 A solid is made by removing a small cone from a large cone to form a frustum.
The large cone has a base radius of 10 cm and a perpendicular height of 24 cm.
The small cone has a base radius of 5 cm and a perpendicular height of 12 cm.



NOT TO SCALE

(a) Calculate the volume of the frustum.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[4]**

(c) The solid frustum is melted down to form solid spheres of radius 3 cm.
Calculate the maximum number of complete spheres that can be made.

..... **[3]**